

P7.1 INTRODUCTION

P7.1.1 Overview

- (a) Building services work shall be provided in the following locations:
 - (i) Digestion Pre-Treatment (WSD11);
 - (ii) Sidestream Nitrogen Removal (WST02).
- (b) Building services shall comply with General Specification G7 Building Services. Reference shall be made to all other Particular and General Specifications.
- (c) The Drawings and Specifications describe notional design for building services and the Specifications provide minimum performance requirements.
- (d) Contractor's design and build scope shall include, but not limited to:
 - (i) Detailed Calculations;
 - (ii) Pipe hydraulic calculations;
 - (iii) Selection of equipment and materials;
 - (iv) Verification of final pressure and voltage drop for services;
 - (v) Verification of effective water storage facilities;
 - (vi) Acoustic, vibration, support, plinth and bracket design for equipment and pipes, ducts, cables and etc.;
 - (vii) Detailed final layout drawings of sprinkler, hose reels and hydraulic calculations of such; and
 - (viii) All the above to be submitted to the S.O. for final approval.
- (e) The Contractor shall engage the appropriate parties and Fire Engineer (FE) for obtaining all permits in relation to engineered smoke control system including but not limited to TFP, FSC, TOP and CSC for the works.
- (f) If required by the Contractor's design, the Contractor shall provide electric traction elevator(s) or lift(s) as per General Specifications and is deemed included in the Tender Sum.

P7.2 AIR-CONDITIONING AND MECHANICAL VENTILATION (ACMV)

P7.2.1 General

- (a) This section specifies the requirements necessary for the provision of fully functional Air-Conditioning and Mechanical Ventilation (ACMV) systems.
- (b) The ACMV System shall be monitored and controlled by the Building Management System (BMS) for normal operation and fire mode operation.

- (c) The ACMV shall comply with the latest energy efficiency standard for building services spelt out under SS530.
- (d) Access platforms, ramps and steps over pipes and trunkings on floor and ladders shall be provided to facilitate necessary passage access to plant locations requiring to be accessed and to safely access all items of plant including equipment, valves and instruments. The platforms, stairways and ladders shall be designed taking into consideration any special access requirement that may be associated with all specific items of plant in question with respect to repair, maintenance, cleaning, greasing/lubricating, draining, replacement, adjustment, dismantling, hoisting operations, assembly, etc. The configuration of the platform, stairway and ladder shall be designed to suit each plant item and location requiring them.
- (e) The Contractor shall appoint a Professional Engineer to carry out the design of all relevant support elements and shall obtain approval from relevant statutory or Government authorities, such as BCA, after liaising with S.O. and Accredited Checker. The cost of design of support system is deemed to have been included in the Contract sum.

P7.2.2 Buildings / Rooms Mechanical Ventilation (MV)

- (a) Mechanical ventilation shall be provided as per Fire Code requirements.
- (b) All non-naturally ventilated escape staircases, if required, shall be provided with MV with emergency power supply to achieve a minimum of four (4) air changes per hour (ACH) under normal ventilation to SS 553 latest edition, and fire mode staircase pressurisation shall maintain a pressure differential of not less than 50 Pa between the pressurised exit staircase and the occupied area when all doors are closed to SCDF Code of Practice for Fire Precautions in Buildings. The MV fans shall be located on the Roof Level and shall be linked to the fire alarm system. Pressure relief damper(s) are to be provided in each staircase to regulate the pressure as per Fire Code. Sufficient airflow at required velocities through door openings as per Fire Code is to be provided.
- (c) Smoke stop lobby, if required, MV system with emergency power supply shall be provided to achieve a minimum of four (4) ACH under normal operation and a minimum of ten (10) ACH under fire mode operations. The MV fans shall be linked to the fire alarm system.
- (d) Redundancy as per Fire Code is to be provided.
- (e) All motorised dampers shall be provided with an external red indicator to indicate when the damper is in its open or close position. The indicator shall align with the ductwork when in the open position.
- (f) MV systems with redundancy fans shall be programmed in the BMS on an equal duty (run time) basis for rotation of duty/standby operation.

- (vii) Disconnect via software programming in lieu of a physical switch as described above is not allowed ; and
- (viii) System shall be tested and commissioned as per General Specifications, Particular Specifications and NFPA 2001, including NFPA2001 Annex C and authority having jurisdiction.

P7.6.4 Fire Hose Reel System

- (a) The Contractor shall provide a fully functional fire hose reel (HR) system complete with 30m hose reel drum, SS304 hose reel cabinet, associated pipework, valves and fittings, etc. as a whole.
- (b) Pipework material for hose reel to be of SS304 complete with SS304 fittings for elbow, tees, etc.
- (c) HR water supply shall be taken from potable water tie-in. Contractor shall liaise and co-ordinate with Contract 1A.

P7.6.5 Dry Riser System

- (a) The Contractor shall provide a fully functional dry riser system complete with dry riser stand pipe, pressure reducing valves, landing valves complete with pressure release valves, 2-way breeching inlets, stand-by hose, associated accessories, etc. as a whole.
- (b) The Contractor shall provide dry riser to each staircase core complete with 2-way breeching inlet at the 1st storey level.
- (c) Earthing shall be provided for each dry riser at the 2-way breeching inlet location.

P7.6.6 Public Address System

- (a) The Contractor shall provide a fully functional Public Address (PA) system complete with amplifiers, pre-amplifiers, IP network audio adaptor, IP network switch, box speakers, cabling, associated accessories, etc. in compliance with the One-Way Emergency Communication System described in Fire Code.
- (b) The Contractor shall terminate the system at a C5A patch panel located in the Non-Process XLV Room. The PA speakers in each building will be linked and connected to the One-Way Voice Emergency Communication System main panel located in the FCC at Industrial Liquid Module Headworks 1 (WIL11) by Contract 5A. The Contractor shall ensure full integration with the existing system (TOA, VX-3000 Series ; Japan).
- (c) The minimum sound level in the area covered shall be at least 65dBA or 10dBA above the ambient noise level, whichever is higher and shall not exceed 105dBA.